

MENA, AR, USA

WMO: 720172

Lat: 34.545N

Lon: 94.203W

Elev: 1079

StdP: 14.13

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 53996

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.0	26.6	6.9	8.2	31.2	11.5	10.3	35.5	20.8	50.1	18.4	49.3	2.5	0	0.379

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	100.3	73.2	98.8	73.3	95.4	74.4	80.7	85.9	79.2	86.2	78.0	86.7	5.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.3	158.6	82.8	77.4	148.4	82.4	75.5	139.0	81.7	45.4	85.0	43.7	83.7	42.4	87.1	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.8	14.0	11.9	13.9	104.1	5.3	3.8	10.0	106.9	6.9	109.1	3.9	111.2	0.1	113.9		
			12.4	81.6	4.8	2.9	8.9	83.7	6.1	85.4	3.4	87.0	-0.1	89.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.6	43.3	47.0	55.3	63.1	70.7	79.9	83.2	82.3	75.1	63.6	53.5	45.4
	DBStd	16.20	10.00	10.08	9.81	8.18	7.28	4.63	4.75	5.56	6.32	8.62	9.37	9.04
	HDD50	755	256	164	55	7	1	0	0	0	4	66	202	
	HDD65	2776	675	507	327	130	32	0	0	4	131	360	610	
	CDD50	5734	48	81	220	401	642	897	1031	1001	754	428	173	58
	CDD65	2280	3	4	27	74	208	447	566	536	308	89	16	2
	CDH74	25841	2	17	173	646	1999	5237	7060	6439	3235	943	85	5
	CDH80	12505	0	0	17	115	613	2557	3881	3553	1485	276	8	0
Wind	WSAvg	4.0	4.9	5.1	5.4	5.4	3.9	3.3	3.1	2.6	2.5	3.2	4.1	4.3
Precipitation	PrecAvg	56.9	3.3	3.7	5.4	5.4	6.7	4.8	4.8	2.7	4.8	5.5	5.4	4.8
	PrecMax	84.6	6.7	9.0	12.8	13.6	18.1	13.9	12.4	6.9	12.0	16.7	13.6	14.3
	PrecMin	31.5	0.3	0.6	1.3	0.4	1.1	0.3	0.2	0.3	0.0	0.0	0.9	0.4
	PrecStd	11.5	1.9	1.8	2.6	2.9	3.5	2.9	3.0	1.5	3.0	4.1	3.3	3.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.4	77.1	81.9	87.6	91.3	99.7	104.3	106.1	99.4	90.3	81.3	73.4
		MCWB	58.6	63.7	64.5	66.5	73.0	71.4	71.3	72.6	71.1	70.9	67.2	64.8
	2%	DB	68.4	72.4	79.0	82.3	89.6	97.0	100.2	101.6	95.3	87.5	76.6	71.5
		MCWB	61.1	63.3	63.0	65.8	72.1	73.4	73.0	73.8	72.1	70.9	65.9	64.6
	5%	DB	65.8	69.8	75.0	80.9	86.2	94.6	98.7	99.0	91.1	82.2	72.7	67.6
		MCWB	61.3	63.1	62.4	65.4	71.1	74.2	73.6	73.8	72.7	68.0	63.7	61.3
	10%	DB	61.5	64.4	72.2	78.6	82.5	90.9	95.2	95.0	89.5	79.4	70.1	62.8
		MCWB	51.1	57.1	61.0	64.9	70.7	74.2	75.0	74.4	72.4	67.3	62.0	54.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.1	70.2	69.5	74.2	80.4	83.3	81.9	81.4	79.4	76.7	73.2	69.9
		MCDB	68.2	71.9	74.1	78.1	84.3	87.8	87.1	86.1	85.0	82.5	76.7	71.7
	2%	WB	64.5	67.2	67.3	71.3	76.9	81.0	80.1	79.6	77.5	74.9	69.6	66.2
		MCDB	66.9	70.9	74.0	78.3	82.8	84.5	87.7	86.0	85.1	82.0	73.1	69.8
	5%	WB	60.4	63.4	65.4	69.6	74.9	78.8	78.9	78.2	76.4	73.1	66.3	61.3
		MCDB	64.5	66.5	73.1	75.7	81.7	85.1	87.8	87.4	84.6	79.7	71.0	64.9
	10%	WB	53.7	59.5	63.2	67.9	73.3	77.2	77.8	77.1	75.1	70.5	63.2	56.4
		MCDB	57.8	64.1	69.7	75.2	80.6	86.2	87.7	87.8	83.5	75.9	67.6	60.6
Mean Daily Temperature Range	5% DB	MDBR	20.1	19.9	21.8	22.2	20.4	21.0	21.7	22.5	23.1	23.9	22.0	19.8
		MCDBR	24.7	24.3	24.0	23.8	21.7	25.0	25.4	26.7	26.7	26.7	23.0	25.8
		MCWBR	17.3	15.9	12.9	11.6	9.0	7.4	6.0	6.3	8.8	12.0	14.5	20.0
	5% WB	MCDBR	20.3	20.1	20.3	17.9	17.8	18.2	19.9	20.8	20.0	19.2	17.2	21.9
	MCWBR	18.3	16.5	13.7	12.2	10.3	9.2	8.1	8.2	8.5	11.4	15.4	19.9	
Clear-Sky Solar Irradiance	taub	0.313	0.318	0.350	0.382	0.411	0.438	0.454	0.453	0.391	0.347	0.325	0.306	
	taud	2.510	2.500	2.432	2.344	2.320	2.278	2.269	2.254	2.396	2.501	2.519	2.543	
	Ebn at Noon	285	296	293	287	278	269	264	262	276	281	278	281	
	Edh at Noon	26	30	34	39	41	43	43	43	35	29	26	24	
All-Sky Solar Radiation	RadAvg	813	937	1297	1609	1756	1989	1966	1820	1552	1196	856	700	
	RadStd	74	136	93	97	165	202	124	134	131	171	105	90	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page